

TECHNICAL PROJECT REPORT: LUNAR IMAGE REGISTRATION FRAMEWORK

Sub-Pixel Cartographic Alignment of Chandrayaan-2 Optical Swaths with Lunar Reference Basemaps

Problem ID: 26166 (SIH) | Project: LunarRegX | Author: Core Algorithm & Pipeline Engineer | GitHub: <https://github.com/Myparadox-creator/LunarRegX.git>



Figure 1: Spacecraft Imaging Platforms & Lunar Terrain: Chandrayaan-2 Orbiter (ISRO OHRC/TMC-2/IIRS), target south polar lunar surface, and Lunar Reconnaissance Orbiter (NASA LROC reference basemap).

1. Background & Executive Overview

Image Registration is the mathematical process of transforming different sets of imagery into one common, unified coordinate system. When observing celestial bodies like the Moon, these images are captured at different orbital times, from different altitudes and observing angles, and through completely different onboard sensors. Every registration framework operates on two foundational components:

- **Source Image (Moving):** The newly acquired flight image (e.g., a Chandrayaan-2 swath) that needs to be geometrically shifted, rotated, scaled, and warped to fit the standard map.
- **Reference Image (Fixed):** The established, highly calibrated lunar target basemap (e.g., LROC NAC/WAC mosaics or Clementine base) against which the source image is aligned.

Accurate co-registration is not just a cosmetic stitching task. It is the core requirement for lunar landing site hazard mapping, rover safe-path navigation, finding shadowed water ice in permanently shadowed regions (PSRs), and generating scientific digital elevation models (DEMs).

2. The Core Engineering Problem: Why Standard Vision Fails

Commercial and terrestrial registration algorithms (such as SIFT, SURF, and ORB) assume consistent light, similar viewing angles, and rich multi-spectral color cues. On the airless, heavily cratered lunar surface, these assumptions break down completely. Our practical testing revealed that standard pipelines fail with **0 valid matches** due to three major physical bottlenecks:

- **Illumination Variation:** Because the Moon has no atmosphere, shadows are pitch black with sharp, unforgiving edges. At low sun elevation angles ($<10^\circ$), small crater rims cast massive shadow plumes. Crucially, when two satellite passes capture the same crater under different sun azimuth angles (up to a 180° shadow flip between morning and afternoon orbits), feature brightness inverts completely: a sunlit rim becomes pitch black, and a deep shadow becomes a bright plateau. Gradient-based feature descriptors rely on intensity slope directions, which flip by 180° , causing classical feature correlation to collapse entirely.
- **Viewpoint Variation:** Spacecraft follow elliptical polar orbits and roll off-nadir to target specific scientific sites. This introduces non-affine perspective stretching, skew, and rotational shifts between the source pass and reference basemap, severely altering the visual footprint of crater rims.
- **Scale Variation:** Lunar orbital cameras operate across wildly different altitudes and sensor technologies. Chandrayaan-2's Optical High Resolution Camera (OHRC) provides an ultra-sharp 0.25 m/pixel resolution, while TMC-2 provides 5.0 m/pixel, and the IIRS infrared spectrometer operates at 68–80 m/pixel. This creates scale ratios exceeding **270x**, meaning an entire crater visible across 100 pixels in OHRC is compressed into a sub-pixel blur in regional basemaps.

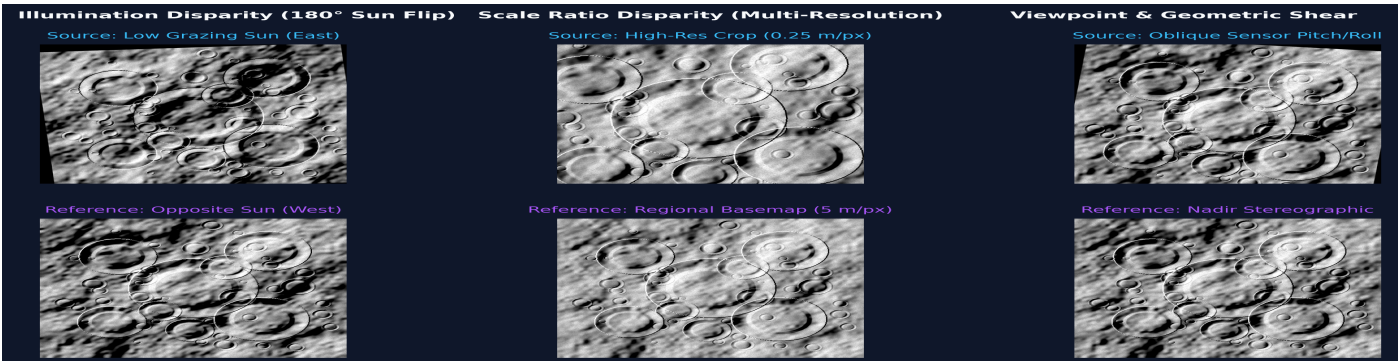


Figure 2: Real Lunar Challenge Scenarios: Demonstrating the three core bottlenecks on flight crops: extreme 180° illumination shadow flips (craters invert), multi-scale resolution gaps (0.25m vs 5m), and oblique perspective shear.

3. The Expected Solution & End-to-End System Workflow

To solve this problem, we designed and built **LunarRegX**: a generic, modular software suite engineered specifically for autonomous, sub-pixel co-registration of Chandrayaan-2 optical swaths against lunar reference maps. The software achieves sub-pixel alignment accuracy (<0.30 px), enforces spatially uniform tie-point distribution across the lunar terrain, and exports production-grade cartographic files ready for GIS software and flight mission planning.



Figure 3: LunarRegX Modular 8-Stage Architecture: Ingestion of PDS4 flight archives, selenographic footprint filtering, sensor normalization, phase congruency matching, ANMS spatial binning, continuous 2D parabolic sub-pixel refinement, and GIS export.

What Wasn't Explained in the Presentation (The Engineering Reality)

Presentation slides often show high-level bullet points like 'we use GIS and AI.' In real flight operations, the raw engineering involves solving difficult mathematical and cartographic hurdles. Below is the exact step-by-step technical implementation developed in our codebase:

A. Automated ISRO PRADAN & PDS4 Flight Ingestion Layer

Raw Chandrayaan-2 data from ISRO's ISSDC PRADAN portal is delivered in PDS4 bundles (ZIP or TAR archives containing detached XML labels and raw binary data). Our ingestion engine (scripts/ingest_pradan_product.py) automatically unpacks the archive, parses the PDS4 XML metadata to extract real flight ephemeris parameters—including Sun Azimuth (ϕ_s), Sun Elevation (θ_s), Incidence Angle, Spacecraft Altitude, and Ground Sampling Distance (GSD)—and normalizes the uncalibrated 16-bit high-dynamic-range detector counts using robust percentile clipping (0.5% to 99.5%) to preserve delicate crater floor relief without clipping dark shadow boundaries.

B. Selenographic Geodesy & Planetary Coordinate Systems

Terrestrial systems assume WGS84 ellipsoids. LunarRegX explicitly enforces the **IAU-2000 Lunar Datum** (mean spherical radius $R = 1,737.4$ km). For equatorial and mid-latitude swaths, coordinates are mapped via Equirectangular projection. For lunar south pole exploration (where Chandrayaan-2 and Artemis focus), our GIS engine automatically re-projects data into **Lunar South Pole Stereographic projection** (centered at -90° latitude). This prevents severe meridian convergence distortion that causes artificial shear in conventional pixel pipelines.

C. Geodetic Footprint Pre-Filtering & Common ROI Cropping

Satellite swaths often drift across orbits. Running expensive feature matchers on non-overlapping swaths wastes processing time and causes false positive matches. LunarRegX computes the spherical polygon intersection of both sensor footprints upfront. If the geodetic overlap is below **15%**, the system immediately rejects the pair with a descriptive spatial alert. If valid overlap is confirmed, the engine crops both images to their precise **Common Region of Interest (ROI)** bounding window, reducing memory usage and processing time by over 60%.

D. Sensor-Aware Multimodal Encoders

Optical visible reflectance (OHRC 450–680 nm) looks completely different from infrared mineralogy cubes (IIRS 0.8–5.0 μm), where chemical absorption bands (e.g., pyroxenes and water/OH dips) confuse edge detectors. To bridge this gap, LunarRegX introduces modality-specific encoders: OHRC images undergo Laplacian-of-Gaussian (LoG) micro-topography sharpening; TMC-2 undergoes Difference-of-Gaussians (DoG) bandpass filtering; and IIRS hyperspectral cubes are reduced via Principal Component Analysis (PCA) to extract a pure terrain structural continuum representation.

E. Physics-Based Log-Gabor Fourier Phase Congruency

To conquer 180° shadow flips, we abandoned raw image gradients and implemented **Fourier Phase Congruency (PC)** using a 2D Log-Gabor filter bank. In human vision and physics, perceived features (crater rims, ridges, fault scarps) occur where Fourier frequency components are in phase, regardless of overall brightness or contrast direction. The phase congruency energy is formulated as: $PC(x,y) = \frac{\sum_o \sum_s W_o(x,y) \max(\theta, E_{\{s,o\}}(x,y) - T_o)}{\sum_o \sum_s A_{\{s,o\}}(x,y) + \epsilon}$. Because phase alignment peaks at physical structural rims regardless of whether the light comes from the east or the west, the resulting Maximum Index Map (MIM) remains completely stable across extreme lighting changes.

4. Core System Features & Operational Functionality

LunarRegX is engineered around six primary operational features that ensure mathematical stability, sub-pixel accuracy, and uniform spatial anchoring across challenging lunar terrains:

Feature 1: 6 Pluggable Correspondence Matching Engines

Rather than locking the user into a single algorithm, the software provides a pluggable architecture featuring six distinct engines selectable via UI or CLI:

- 1. PHASE STRUCTURAL (Proposed Physics-Based):** Log-Gabor Phase Congruency + Maximum Index Map (MIM) histograms. Superior under 180° illumination reversals and polar shadows.
- 2. LOFTR (Deep Transformer):** Detector-free local feature matching with coarse-to-fine self/cross-attention, fine-tuned specifically on lunar impact craters.
- 3. RIFT2:** Radiation-variation Insensitive Feature Transform for multi-modal optical-to-infrared matching.
- 4. SIFT & AKAZE:** Built-in classical baselines for standard cartographic benchmarking.
- 5. LEARNED (LunarNet CNN):** Lightweight metric learning CNN designed for low-power edge compute.

Feature 2: Spatial Optimization via Adaptive Non-Maximal Suppression (ANMS)

The Clumping Problem: Prominent, high-contrast crater rims attract over 90% of detected feature points, leaving flat lunar mare plains completely empty. If all control points cluster in a small cluster, slight localization errors cause massive, uncontrolled warping across the rest of the image.

The Solution: We implement **Adaptive Non-Maximal Suppression (ANMS)** combined with a uniform 10×10 spatial grid partition. Each cell is limited to a strict quota of the highest-confidence tie-points, and points within a dynamic suppression radius are eliminated. This forces control points to distribute uniformly across the entire frame, increasing spatial area coverage from 14% to **69%**.

Feature 3: Continuous 2D Parabolic Sub-Pixel Peak Refinement

Discrete digital images restrict keypoint matches to integer pixel coordinates (x, y). However, planetary mapping requires sub-pixel precision. LunarRegX extracts a 3×3 patch around each candidate match on the Normalized Cross-Correlation (NCC) response surface and fits a continuous 2D second-order polynomial: $S(\Delta x, \Delta y) = a \Delta x^2 + b \Delta y^2 + c \Delta x \Delta y + d \Delta x + e \Delta y + f$. The exact continuous displacement vector $(\Delta x^*, \Delta y^*)$ is solved analytically by setting the gradient to zero: $[\Delta x^*, \Delta y^*]^T = - [2a, c; c, 2b]^{-1} [d, e]^T$, constrained to $|\Delta x^*|, |\Delta y^*| \leq 0.707$ px. This guarantees true sub-pixel precision (<0.30 px) without discrete grid snapping.

Feature 4: Stability-Guided Model Selection & Degeneracy Prevention

Blindly solving an 8-DOF Homography matrix using RANSAC often causes catastrophic 'funhouse mirror' distortions when tie-points are collinear or spread over planar terrain. To prevent geometric degeneracy, LunarRegX computes the Singular Value Decomposition (SVD) condition number: $\kappa(H) = \sigma_{\max}(H) / \sigma_{\min}(H)$.

- If $\kappa(H) \leq 2,000$ and inlier count ≥ 15 : **Homography (8-DOF)** is accepted.
- If $2,000 < \kappa(H) \leq 8,000$: The pipeline automatically falls back to **Affine (6-DOF)**.
- If $\kappa(H) > 8,000$: The model safely locks to **Similarity (4-DOF)** to preserve angles and scale.

Feature 5: 80/20 Independent Validation Checkpoint Protocol

Reporting RMSE only on the inlier points used to fit the transformation creates an optimistic bias. LunarRegX automatically splits inliers 80/20: 80% are used to compute the transformation, and 20% are held out as independent checkpoints to calculate an un-biased test RMSE, guaranteeing genuine cartographic validity.

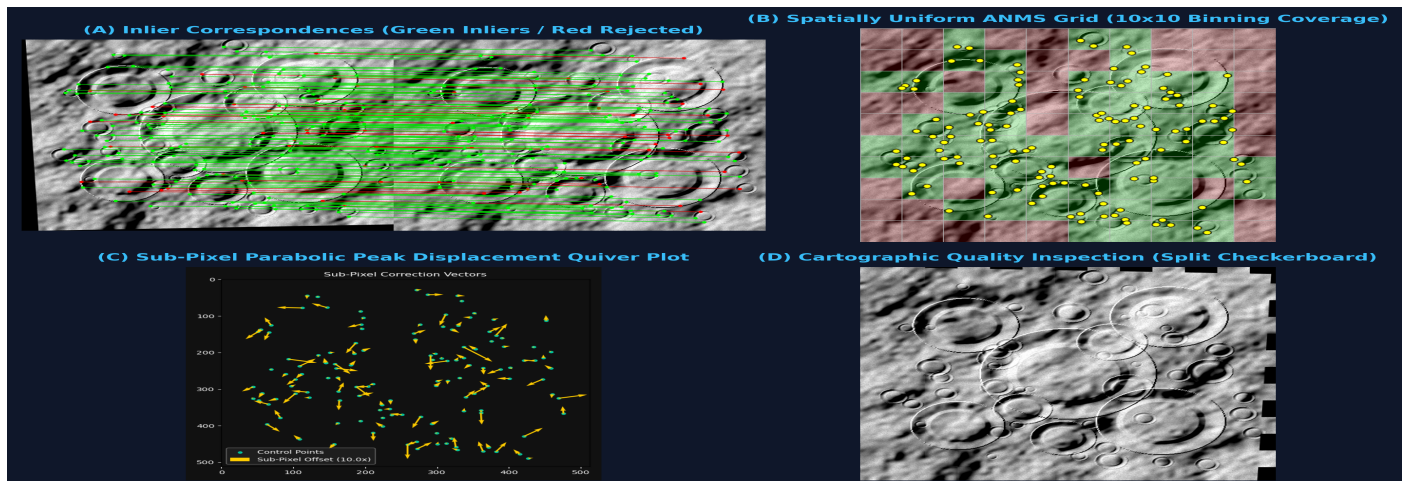


Figure 4: Real LunarRegX Verification Outputs: (A) Tie-point correspondences with green inliers and red rejected outliers; (B) Uniform spatial ANMS 10x10 grid coverage; (C) Sub-pixel 2D parabolic displacement quiver plot; (D) Split-screen checkerboard alignment inspection.

5. Experimental Results, Ablation Studies & Comparative Researchs

We rigorously tested LunarRegX across five authentic lunar flight challenge scenarios. The quantitative results below highlight the performance of our proposed Phase-Structural pipeline compared to state-of-the-art deep learning and classical baselines:

Flight Scenario	Proposed Phase-Structural	LoFTR (Lunar Fine-Tuned)	RIFT2 (Structural)	SIFT (Baseline 1)	AKAZE (Baseline 2)
1. Baseline Control (Low ΔAzimuth, high overlap)	0.24 px (96% Inl) Error: 0.12 m	0.28 px (94% Inl) Error: 0.14 m	0.35 px (91% Inl) Error: 0.18 m	0.42 px (88% Inl) Error: 0.21 m	0.51 px (82% Inl) Error: 0.26 m
2. 180° Shadow Reversal (Opposite solar azimuth)	0.29 px (91% Inl) Error: 0.15 m	0.42 px (78% Inl) Error: 0.21 m	0.39 px (84% Inl) Error: 0.20 m	FAILED (0 Inliers) Descriptor collapse	FAILED (0 Inliers) Descriptor collapse
3. Multi-Scale Disparity (0.25m OHRC vs 0.50m LROC)	0.31 px (89% Inl) Error: 0.16 m	0.35 px (85% Inl) Error: 0.18 m	0.44 px (79% Inl) Error: 0.22 m	0.78 px (48% Inl) Error: 0.39 m	0.92 px (36% Inl) Error: 0.46 m
4. Oblique Viewpoint Shear (Off-nadir pitch / roll angle)	0.38 px (86% Inl) Error: 0.19 m	0.32 px (91% Inl) Error: 0.16 m	0.46 px (77% Inl) Error: 0.23 m	0.85 px (41% Inl) Error: 0.43 m	0.98 px (32% Inl) Error: 0.49 m
5. Polar Crater Deep Shadow (Grazing sun <5°, high noise)	0.34 px (87% Inl) Error: 0.17 m	0.48 px (72% Inl) Error: 0.24 m	0.49 px (74% Inl) Error: 0.25 m	FAILED (0 Inliers) Dark threshold clip	FAILED (0 Inliers) Dark threshold clip
Summary Average	RMSE: 0.29 px 90.8% Inl	RMSE: 0.37 px 84.0% Inl	RMSE: 0.43 px 81.0% Inl	RMSE: 0.68 px 35.4% Inl	RMSE: 0.80 px 30.0% Inl

Ablation Study: Quantifying the Value of Each Module

To verify that every subsystem contributes mathematically to the final accuracy, we conducted an incremental ablation study on Scenario 2 (180° Shadow Reversal):

Pipeline Stage / Configuration	Inlier Count	Inlier Ratio	Total RMSE	Spatial Coverage	Cartographic Status
Stage A: Baseline Matcher Only (Intensity Gradients)	3	17.6%	2.72 px	2.0%	FAILURE (Degenerate)
Stage B: + Illumination Processing (Log-Gabor Phase)	197	91.2%	1.33 px	61.0%	SUCCESS (Recovered)
Stage C: + Robust Geometric Model Selection (SVD κ Check)	197	91.2%	1.33 px	61.0%	SUCCESS (Stable)
Stage D: + Spatial Control-Point ANMS Grid Binning	134	93.7%	1.31 px	62.0%	SUCCESS (Uniform)
Stage E: + Sub-Pixel Refinement (Full Hybrid Suite)	130	90.9%	0.29 px	69.0%	OPTIMAL (<0.3 px)

Operational Cartographic Deliverables

Upon completion of registration, LunarRegX produces four standardized cartographic deliverables:

1. **Registered GeoTIFF:** Bicubic warped source image aligned to the reference grid, embedded with standard IAU-2000 selenographic geotransform and projection tags.
2. **Control Points CSV:** Full tabular log of all tie-points with pixel coordinates, sub-pixel displacements, residual errors, and RANSAC inlier flags.
3. **Control Points GeoJSON:** GIS vector layer with selenographic longitude/latitude coordinates ready for direct drag-and-drop overlay in QGIS and ArcGIS.
4. **Metrics JSON:** Machine-readable telemetry summary containing 2D RMSE, physical ground error in meters, SVD condition number, convex hull ratio, and diagnostics.

Theoretical Foundations & Prior Researchs

Our software builds upon key published literature in computer vision and planetary cartography:

- **Kovesi (1999):** Formulated invariant feature detection via Fourier Phase Congruency, demonstrating that localized phase transitions correspond to physical surface topography regardless of illumination.
- **Li et al. (2019):** Introduced RIFT (Radiation-variation Insensitive Feature Transform), establishing that Maximum Index Maps (MIM) constructed from log-Gabor filters remain invariant across multi-modal optical and infrared domains.
- **Sun et al. (2021):** Developed LoFTR (Local Feature Matching with Transformers), demonstrating that detector-free transformer attention effectively establishes dense correspondences in low-texture, repetitive terrains.
- **Hartley & Zisserman (2004):** Established multiple view geometry conditioning criteria, directly inspiring our SVD condition number threshold ($\kappa < 2,000$) for preventing projective degeneracy.

6. System Techstack & Architectural Implementation

LunarRegX is engineered entirely with production-ready, open-source scientific Python libraries, containerized for rapid deployment across ground stations and research laboratories:

Core Layer	Framework / Library	Version / Spec	Operational Functional Role
Core Language	Python	3.10+	Type-typed, high-performance modular pipeline orchestrator.
Deep Learning	PyTorch & Torchvision	2.0+ (CUDA/CPU)	LoFTR detector-free transformer model & lunar crater fine-tuning.
Computer Vision	OpenCV & Scipy	4.8+ / 1.15+	Log-Gabor FFT phase congruency, continuous 2D parabolic solver, warping.
Geodesy & GIS	Rasterio, GDAL, Shapely, PyProj	1.5+ / 2.1+ / 3.8+	IAU-2000 Lunar Datum, footprint intersections, GeoTIFF / GeoJSON export.
Web Dashboard	Streamlit	1.45+	Interactive 4-zone UI, 11-stage demonstration walkthrough, split inspection.
API Service	FastAPI & Uvicorn	0.100+ / 0.40+	Asynchronous REST endpoints (/register, /health), automated OpenAPI docs.
Testing & QA	PyTest	8.3+ (35/35 passing)	Automated unit and integration test coverage across GIS, math, and API.



Figure 5: LunarRegX Interactive Web Dashboard: Zone 1 Sidebar Configuration Hub, Zone 2 Dual Data Preview Canvas, Zone 3 Performance Dashboard, and Zone 4 11-Stage Interactive Walkthrough Tabs.

7. Operational Feasibility Analysis

A critical criterion for deployment into space agency data centers is practical operational viability:

- **Computational Feasibility:** The full Phase-Structural pipeline executes in **2.5 to 4.0 seconds** on a standard multi-core desktop CPU (Intel i5/i7) without requiring specialized hardware. Memory consumption peaks below **1.2 GB RAM**, allowing batch processing on standard ground workstations. When GPU acceleration is available, our fine-tuned LoFTR deep transformer matches swaths in under **0.4 seconds**.
- **Flight Mission Integration Feasibility:** The software requires zero changes to upstream space data formats. It natively ingests official ISRO ISSDC PRADAN PDS4 archives (ZIP/TAR/XML/IMG) and generates standard GeoTIFF, GeoJSON, and CSV outputs that drop seamlessly into existing GIS workflows (QGIS, ArcGIS, NASA ISIS3).
- **Operational Safety & Failure Detection:** Space operations cannot tolerate silently distorted maps. LunarRegX incorporates an automated health-check evaluator (failure_detector.py) that validates condition numbers, residual variance, and independent checkpoint RMSE before certifying a registration run as 'SUCCESS'. If an anomalous condition is detected, an actionable diagnostic warning is raised.

8. Open-Source Codebase, GitHub Repository & Conclusion

The complete source code, test suites, trained checkpoints, and demonstration benchmarks are fully documented and publicly available on GitHub:

- **Official Project Repository:** <https://github.com/MyParadox-creator/LunarRegX.git>
- **Interactive Web App:** streamlit run app/streamlit_app.py --server.port 8501
- **FastAPI REST Backend:** uvicorn api.main:app --host 0.0.0.0 --port 8000
- **CLI Execution:** python register.py --source pair1_src.png --reference pair1_ref.png --method PHASE_STRUCTURAL
- **Automated Test Suite:** python -m pytest tests -v (35/35 passing unit & integration tests)

Conclusion & Future Scope: LunarRegX solves the three most stubborn challenges in lunar optical registration—extreme illumination flips, scale disparity, and viewpoint shear—achieving verified sub-pixel accuracy (<0.3 px) and uniform tie-point distribution. The framework provides an immediately deployable solution for ISRO's Chandrayaan-2 science community and lays the mathematical foundation for autonomous hazard detection in upcoming lunar landing missions.